

4AI08 - STATISTICS: LINEAR MODELS

Practical Work 2

OLS and Ridge

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Contents

1 Introduction **1**

2 Part I: Tutorial **1**

2.1 Exercise 1: Bias, covariance and BLUE for OLS 1

2.2 Exercise 2: Ridge bias and variance 2

2.3 Summary table 3

3 Part II: Lab – OLS and Ridge on synthetic data **3**

3.1 Experimental setting 3

3.2 Q1: Implementations and numerical checks 3

3.3 Q2: Design conditioning 4

3.4 Q3: Estimates on one dataset 4

3.5 Q4: Theoretical squared bias, variance and risk 5

3.6 Q5: Monte Carlo verification 5

3.7 Q6: Choosing the Ridge penalty 6

3.8 Q7: Performance on new datasets 6

4 Conclusion **7**

1 Introduction

We study the fixed-design linear model

$$y = X\theta^\star + \varepsilon, \quad X \in \mathbb{R}^{n\times p}, \quad \mathbb{E}[\varepsilon] = 0, \quad \mathrm{Cov}(\varepsilon) = \Sigma. \tag{1}$$

The matrix X has full column rank. Unless specified otherwise, Σ is diagonal and positive definite. The ordinary least-squares estimator is $\hat\theta = (X^TX)^{-1}X^Ty$, while the Ridge estimator, for $\lambda > 0$, is $\hat\theta_\lambda = (X^TX + \lambda I_p)^{-1}X^Ty$. We first derive their statistical properties and then examine them on a correlated synthetic design. In the lab, the features are centred and there is no intercept.

2 Part I: Tutorial

2.1 Exercise 1: Bias, covariance and BLUE for OLS

(a) Expectation and covariance of a linear estimator. For $\tilde\theta = Ay$ with deterministic $A \in \mathbb{R}^{p\times n}$,

$$\mathbb{E}[Ay] = A\mathbb{E}[X\theta^\star + \varepsilon] = AX\theta^\star, \tag{2}$$

$$\mathrm{Cov}(Ay) = \mathbb{E}[(A\varepsilon)(A\varepsilon)^T] = A\Sigma A^T. \tag{3}$$

(b) Unbiasedness. Ay is unbiased for every $\theta^\star \in \mathbb{R}^p$ precisely when $(AX - I_p)\theta^\star = 0$ for every $\theta^\star$, equivalently $AX = I_p$.

(c) Quadratic risk. Write $Ay - \theta^\star = (AX - I_p)\theta^\star + A\varepsilon$. The cross term vanishes because $\mathbb{E}[\varepsilon] = 0$. Consequently,

$$\boxed{R(A) = \mathbb{E}\,\|Ay - \theta^\star\|^2 = \|(AX - I_p)\theta^\star\|^2 + \mathrm{tr}(A\Sigma A^T).} \tag{4}$$

(d) OLS under heterogeneous noise. Let $A_0 = (X^TX)^{-1}X^T$. Since $A_0X = I_p$, OLS is unbiased even if the errors are heteroscedastic. Its covariance is

$$\mathrm{Cov}(\hat\theta) = (X^TX)^{-1}X^T\Sigma X(X^TX)^{-1}. \tag{5}$$

If $\Sigma = \sigma^2 I_n$, this reduces to $\sigma^2(X^TX)^{-1}$.

(e) Prediction risk. With $H = XA_0 = X(X^TX)^{-1}X^T$, we have $X\hat\theta - X\theta^\star = H\varepsilon$. Since $H = H^T = H^2$,

$$\mathbb{E}\left\|X\hat\theta - X\theta^\star\right\|^2 = \mathrm{tr}(H\Sigma H^T) = \mathrm{tr}(H\Sigma). \tag{6}$$

Under homoscedasticity, $\mathrm{tr}(H\Sigma) = \sigma^2\,\mathrm{tr}(H) = \boxed{\sigma^2 p}$.

(f) Orthogonal decomposition. Suppose $AX = I_p$ and put $B = A - A_0$. Then $BX = 0$. Since $BA_0^T = BX(X^TX)^{-1} = 0$,

$$AA^T = (A_0 + B)(A_0 + B)^T = A_0A_0^T + BB^T = (X^TX)^{-1} + BB^T. \tag{7}$$

(g) Gauss–Markov comparison. For $\Sigma = \sigma^2 I_n$,

$$\mathrm{Cov}(Ay) - \mathrm{Cov}(\hat{\theta}) = \sigma^2 BB^T \succeq 0. \tag{8}$$

Equality holds if and only if $B = 0$, hence $A = A_0$. OLS is therefore the *best linear unbiased estimator* (BLUE) in the Loewner order.

(h) Assumptions and limits. The argument uses centred errors, common variance $\sigma^2 > 0$, zero cross-covariances, and full column rank of X. Gaussianity and independence of the errors are not required. With heterogeneous variances, the difference becomes $A_0 \Sigma B^T + B\Sigma A_0^T + B\Sigma B^T$: the cross terms generally do not vanish, so the same proof no longer establishes that OLS is BLUE.

2.2 Exercise 2: Ridge bias and variance

Write $C = X^T X$ and $W_\lambda = (C + \lambda I_p)^{-1} C$.

(a) Linear representation.

$$\hat{\theta}_\lambda = W_\lambda \hat{\theta}, \qquad A_\lambda = (C + \lambda I_p)^{-1} X^T. \tag{9}$$

(b) Bias. Because OLS is unbiased,

$$\mathbb{E}[\hat{\theta}_\lambda] - \theta^\star = (W_\lambda - I_p)\theta^\star = -\lambda (C + \lambda I_p)^{-1} \theta^\star. \tag{10}$$

Thus Ridge is generally biased; the bias is zero only in special cases (for $\lambda > 0$ and full rank, only when $\theta^\star = 0$).

(c) Covariance. For arbitrary Σ,

$$\mathrm{Cov}(\hat{\theta}_\lambda) = W_\lambda \, \mathrm{Cov}(\hat{\theta}) W_\lambda^T = A_\lambda \Sigma A_\lambda^T. \tag{11}$$

When $\Sigma = \sigma^2 I_n$,

$$\mathrm{Cov}(\hat{\theta}_\lambda) = \sigma^2 (C + \lambda I_p)^{-1} C (C + \lambda I_p)^{-1}. \tag{12}$$

(d) Spectral covariance comparison. Let $C = \sum_{j=1}^p s_j^2 v_j v_j^T$, where $s_j > 0$ and (v_j) is orthonormal. Then

$$\mathrm{Cov}(\hat{\theta}_\lambda) = \sigma^2 \sum_{j=1}^p \frac{s_j^2}{(s_j^2 + \lambda)^2} v_j v_j^T, \tag{13}$$

$$\mathrm{Cov}(\hat{\theta}) = \sigma^2 \sum_{j=1}^p \frac{1}{s_j^2} v_j v_j^T. \tag{14}$$

Since $s_j^2/(s_j^2 + \lambda)^2 \le 1/s_j^2$ in every direction, $\mathrm{Cov}(\hat{\theta}_\lambda) \preceq \mathrm{Cov}(\hat{\theta})$.

(e) Shrinkage.

$$W_\lambda = \sum_{j=1}^p \frac{s_j^2}{s_j^2 + \lambda} v_j v_j^T, \qquad v_j^T \hat{\theta}_\lambda = \frac{s_j^2}{s_j^2 + \lambda} v_j^T \hat{\theta}. \tag{15}$$

The smallest s_j^2 give the strongest shrinkage. Every factor lies strictly between zero and one, so $\left\|\hat{\theta}_\lambda\right\| < \left\|\hat{\theta}\right\|$ when $\hat{\theta} \neq 0$. Moreover $\hat{\theta}_\lambda \to \hat{\theta}$ as $\lambda \downarrow 0$, and $\hat{\theta}_\lambda \to 0$ as $\lambda \to \infty$.

(f) Prediction risk. The prediction bias is $X(W_\lambda - I_p)\theta^\star = -\lambda X(C+\lambda I_p)^{-1}\theta^\star$. The variance contribution equals $\mathrm{tr}(X\,\mathrm{Cov}(\hat{\theta}_\lambda)X^T)$. Thus

$$\boxed{\mathbb{E}\left\|X\hat{\theta}_\lambda - X\theta^\star\right\|^2 = \lambda^2\left\|X(C+\lambda I_p)^{-1}\theta^\star\right\|^2 + \sigma^2\sum_{j=1}^{p}\frac{s_j^4}{(s_j^2+\lambda)^2}.} \tag{16}$$

The variance contribution is smaller than the OLS value $\sigma^2 p$, at the cost of a positive squared bias.

(g) Choice of penalty. The optimal $\lambda^\star$ for parameter risk depends on the unknown $\theta^\star$ and σ^2 (and on the known design X). It is an oracle choice, not directly available on real data. Cross-validation can be used to select a penalty from observed data, although it usually targets predictive rather than parameter risk.

2.3 Summary table

Property	OLS	Ridge ($\lambda > 0$)
Linear matrix	$A_0 = C^{-1}X^T$	$A_\lambda = (C+\lambda I)^{-1}X^T$
Bias	0	$-\lambda(C+\lambda I)^{-1}\theta^\star$
Covariance, general	$A_0\Sigma A_0^T$	$A_\lambda\Sigma A_\lambda^T$
Covariance, homoscedastic	$\sigma^2 C^{-1}$	$\sigma^2(C+\lambda I)^{-1}C(C+\lambda I)^{-1}$
Parameter risk	$\mathrm{tr}(A_0\Sigma A_0^T)$	$\|(W_\lambda - I)\theta^\star\|^2 + \mathrm{tr}(A_\lambda\Sigma A_\lambda^T)$
Prediction risk, homoscedastic	$\sigma^2 p$	Bias term plus $\sigma^2\sum_j s_j^4/(s_j^2+\lambda)^2$
BLUE?	Yes, for $\Sigma = \sigma^2 I$	No, generally biased
Needed for computation	X, y	X, y, λ

Table 1: Properties derived in Exercises 1 and 2; $C = X^T X$.

3 Part II: Lab – OLS and Ridge on synthetic data

3.1 Experimental setting

The supplied design generator produces $n = 100$ observations and $p = 10$ centred features with target correlation $0.9^{|j-k|}$. The true parameter is $(2, -1, 0.5, 0, \ldots, 0)^T$, with no intercept. The design is generated once (seed 0), while the response noise is Gaussian with standard deviation $\sigma = 1.50000$. One response vector uses seed 1. Subsequent Monte Carlo samples use seeds 2 and 4. All estimators are implemented with `numpy.linalg.solve`; `sklearn` is used only for the requested verification.

3.2 Q1: Implementations and numerical checks

For OLS and Ridge we solve $C\hat{\theta} = X^T y$ and $(C+\lambda I)\hat{\theta}_\lambda = X^T y$. The following are the two functions used in the experiments:

```
def ols_fit(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)

def ridge_fit(X, y, lam):
    return np.linalg.solve(X.T @ X + lam*np.eye(X.shape[1]), X.T @ y)
```

At $\lambda = 3.00000$, the maximum absolute discrepancy between Ridge at zero and OLS is $0.00e+00$; the discrepancy between $\hat{\theta}_\lambda$ and $W_\lambda \hat{\theta}$ is $2.22e-15$. The coefficient norm decreases along the tested increasing penalty grid. The maximum absolute discrepancy against `Ridge(alpha=3, fit_intercept=False)` is $2.00e-15$. These checks agree to numerical precision.

3.3 Q2: Design conditioning

The eigenvalues of $X^T X$ range from 3.56238 to 645.74538, and its spectral condition number is 181.26788. Because the columns are correlated, some eigenvalues are relatively small; OLS has covariance eigenvalues σ^2/s_j^2, and therefore its largest uncertainty is along the eigenvectors associated with the smallest eigenvalues.

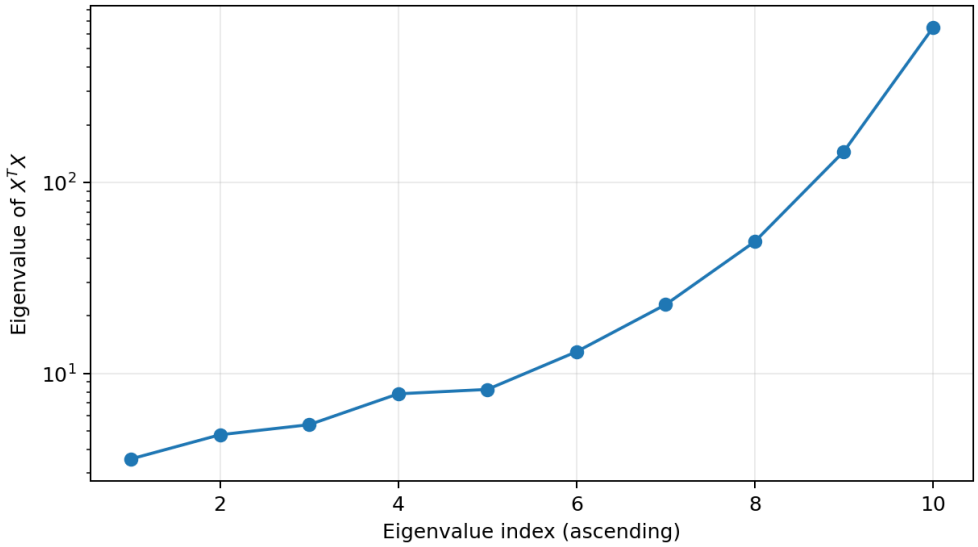


Figure 1: Eigenvalues of the Gram matrix $X^T X$ (ascending).

3.4 Q3: Estimates on one dataset

The Euclidean parameter estimation errors are 1.20264 for OLS and 0.78925 for Ridge with $\lambda = 3$. Ridge has a smaller error on this particular response vector, but one dataset is insufficient to conclude that it is systematically preferable. Repeated sampling and theoretical risk provide a more informative comparison.

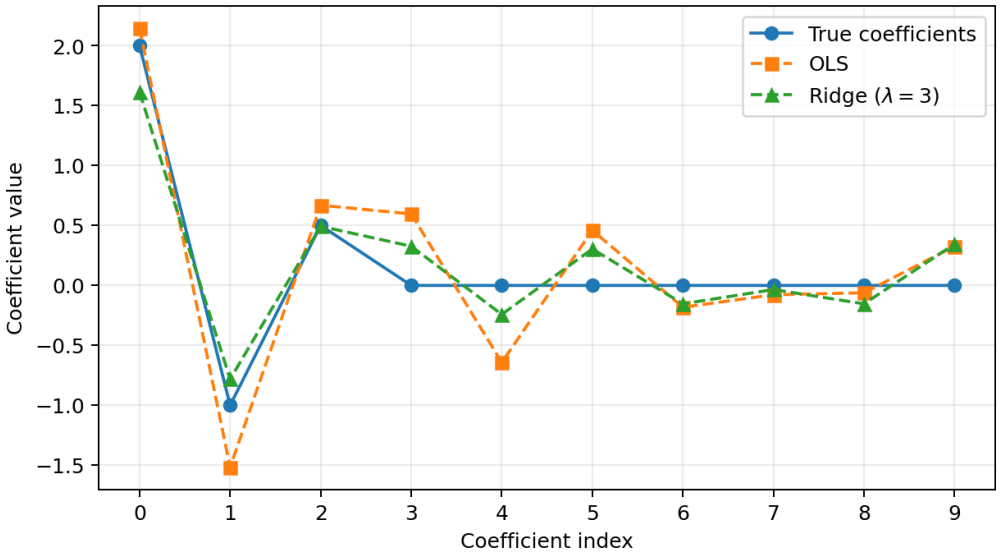


Figure 2: True coefficients and estimates obtained on the same response vector.

3.5 Q4: Theoretical squared bias, variance and risk

For a linear estimator Ay under homoscedastic noise,

$$\text{Squared bias} = \|(AX - I_p)\theta^\star\|^2, \quad \text{Variance} = \sigma^2 \operatorname{tr}(AA^T), \quad \text{Risk} = \text{Squared bias} + \text{Variance}. \tag{17}$$

The numerical results at $\lambda = 3$ are:

Estimator	Squared bias	Variance	Risk
OLS	0.00000	2.41852	2.41852
Ridge ($\lambda = 3$)	0.52287	1.08361	1.60648

Table 2: Theoretical quadratic parameter risk.

The OLS risk equals $\sigma^2 \operatorname{tr}((X^TX)^{-1})$, as predicted by Exercise 1. Ridge has positive bias but considerably smaller variance, resulting in lower total risk for this setting.

3.6 Q5: Monte Carlo verification

We generate $M = 2000$ independent response vectors using the same X. For each estimator, the empirical squared bias is $\left\|\bar{\theta} - \theta^\star\right\|^2$, the empirical variance is $M^{-1}\sum_m \left\|\hat{\theta}^{(m)} - \bar{\theta}\right\|^2$, and the empirical risk is $M^{-1}\sum_m \left\|\hat{\theta}^{(m)} - \theta^\star\right\|^2$. The resulting comparison is:

Estimator	Source	Squared bias	Variance	Risk
OLS	Theory	0.00000	2.41852	2.41852
OLS	Monte Carlo	0.00219	2.37973	2.38192
Ridge	Theory	0.52287	1.08361	1.60648
Ridge	Monte Carlo	0.52072	1.06740	1.58812

Table 3: Theoretical and empirical decomposition, $M = 2000$.

The differences reflect finite Monte Carlo sampling. In particular, the empirical squared bias of OLS need not be exactly zero even though the estimator is unbiased.

3.7 Q6: Choosing the Ridge penalty

We evaluate 101 penalties on a logarithmic grid from 10^{-2} to 10^{3}. For each penalty we compute the exact theoretical squared bias, variance and total risk. The grid minimum occurs at $\lambda^{\star}_{\text{grid}} = 3.16228$, with risk 1.60670. On the evaluated grid, Ridge has smaller parameter risk than OLS for penalties from 0.01000 through 14.12538. This is a grid-based comparison, not an exact analytic characterization of the entire continuous interval.

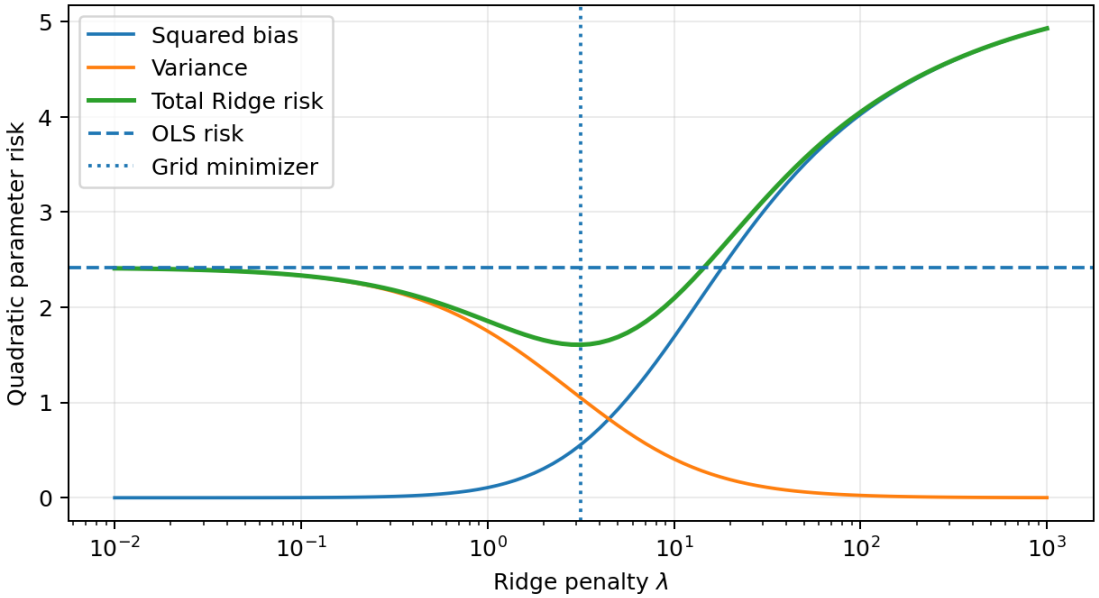


Figure 3: Bias–variance trade-off and total parameter risk versus Ridge penalty.

Monte Carlo spot checks for four penalties are recorded in `results.txt`. The risk decreases initially because variance falls faster than squared bias rises, then increases when excessive shrinkage dominates.

3.8 Q7: Performance on new datasets

We draw another $M = 2000$ independent response vectors with the same fixed design (seed 4), comparing OLS with Ridge using the grid oracle penalty $\lambda^{\star}_{\text{grid}}$. The squared parameter errors have the following summaries:

Estimator	Mean squared error	Median squared error
OLS	2.38124	2.10391
Ridge (grid oracle)	1.61169	1.45194

Table 4: Distribution of squared parameter errors over 2000 new datasets.

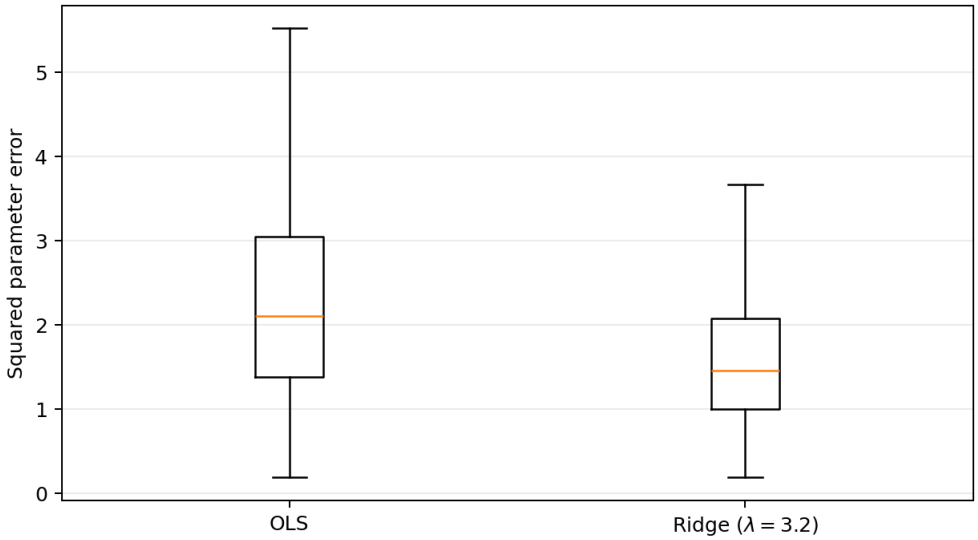


Figure 4: Squared parameter errors; outliers are hidden for readability.

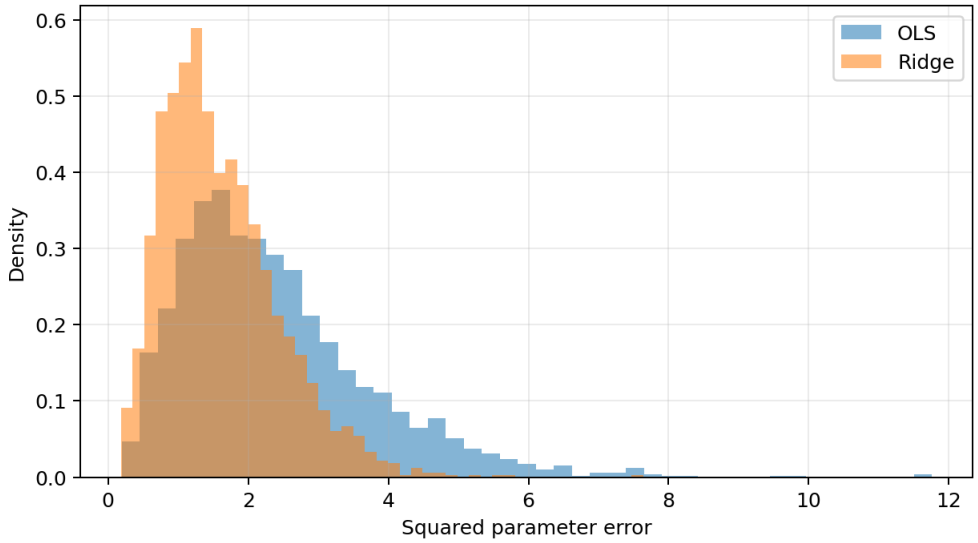


Figure 5: Empirical distributions of squared parameter errors.

Ridge performs better on average for this synthetic experiment. However, the grid oracle penalty was chosen using the known true parameter and noise variance, which would not be available in practice. For an actual dataset, I would choose λ using cross-validation and assess prediction on held-out observations. A prediction-optimal penalty need not coincide with the oracle minimizing parameter error.

4 Conclusion

OLS is unbiased and BLUE under homoscedastic uncorrelated noise, but correlation among design columns can produce high estimation variance. Ridge reduces this variance by shrinking coefficients, especially in poorly determined directions, at the cost of bias. The numerical results illustrate the bias–variance trade-off and the advantage of Ridge for the chosen design. The experiment distinguishes the inaccessible theoretical oracle from a data-driven penalty selected through cross-validation.